

EDUCATION

Texas A&M University, College Station, Texas
Bachelor of Science in Interdisciplinary Engineering
Focus: Biomedical Engineering and Computer Science

Expected Graduation: May 2028
GPA: 3.71

- Engineering Honors (Top 10% of Engineering cohort)

EXPERIENCE

Tabor Research Group

May 2025 – Present

Undergraduate Researcher

- Accelerated the discovery of novel metastable photoacids by computationally filtering **11,000,000+ molecules** with **SMARTS-based structural screening**, reducing manual effort by **over 90%** and collecting valid structural candidates for downstream analysis
- Engineered a generalized **reaction SMARTS pattern** capable of simulating ring isomerization reactions across **300,000+ structurally distinct** photoacid candidate systems
- Delivered slide-based research presentations to a lab group of 15+ graduate students, postdocs, and Professor Tabor, effectively communicating project progress and computational results
- Leveraged **pre-trained machine learning models** through **Python-driven automation** to predict pKa values and integrate results into High-Performance Computing (HPC) workflow

Society of Asian Scientists and Engineers (SASE)

August 2024 – Present

Science Director

- Elected Science Director for TAMU's SASE chapter, sponsored by companies such as Chevron, Boeing, Capital One, and Lockheed Martin, to oversee the planning of **5+ science-related events** per semester for **200+ members**
- Managed an allocated budget and directed a **14-member** committee to organize engaging events including research workshops, outreach events, and academic fair representation to promote undergraduate involvement in research

PROJECTS

Strive (ACC)

September 2024 – May 2025

Project Member

- Collaborated with a 17-member team to develop Strive, a mobile app for class scheduling built with JavaScript, React Native, and Firebase
- Contributed to UI design using Figma and provided usability feedback to enhance app functionality and user experience

TECHNICAL SKILLS

Languages: Python, C++, HTML, JavaScript, CSS, SMILES/SMARTS

Frameworks & Libraries: React, RDKit, pandas, NumPy, SciPy, matplotlib, PyTorch

Software: Gaussian 16, CREST, ChemDraw, SOLIDWORKS, AutoCAD 2D/3D, Onshape, Figma, Excel

Developer Tools: Git/GitHub, VS Code, PyCharm, SLURM, HPC Workflows

Relevant Coursework: Data Structures & Algorithms, Biomedical Computing & Design, Program Design & Concepts, Discrete Structures for Computing, Biostatistics & Data Visualization, Organic Chemistry I & II, Physiology for Bioengineers